AQUACARBON DC

High-Level Product Overview | **AI Builder Cup — Sustainability & Social Impact**

One-line vision: Help data-center teams compare cooling strategies by making the trade-offs between water, electricity, carbon and cost visible—without compromising thermal safety or operational reliability.

1. The problem

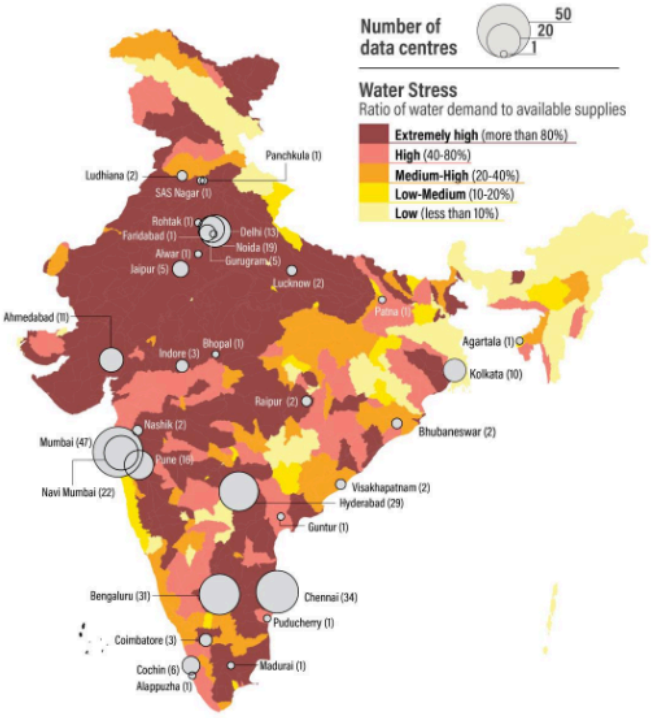
Data centers must continuously remove heat from IT equipment, but cooling strategies have different environmental and operational consequences. A configuration that reduces onsite water use may require more electricity; additional electricity can increase operating cost, carbon emissions and indirect water consumption associated with power generation. Optimizing only Power Usage Effectiveness (PUE) can therefore hide important trade-offs. The best option depends on local climate, water availability, workload, grid emissions and engineering constraints.

2. Why this matters

The International Energy Agency estimated that data centers consumed about 415 TWh of electricity globally in 2024 and projects around 945 TWh by 2030 in its base case. WRI India reported in May 2026 that more than half of India's data centers are located in water-stressed regions, with roughly 75% concentrated across Maharashtra, Tamil Nadu, Karnataka, Telangana and Uttar Pradesh. These figures establish the relevance of the problem; they are not inputs for predicting any individual facility's consumption.

More than half of India's data centres are located in water-stressed regions

Five states—Maharashtra, Tamil Nadu, Karnataka, Telangana, Uttar Pradesh—account for 75% of the country's 278 data centres, which require large amounts of water for cooling.



Note: Water stress measures the ratio of total water demand to available renewable surface and groundwater supplies. Water demand include domestic, industrial, irrigation and livestock uses. Higher values indicate more competition among users.

Source: **Datacentermap and WRI Aqueduct**, accessed on April 12, 2026. Analysis and visualisation: Rajit Sengupta
Disclaimer: The map is for illustrative purposes and does not imply the expression of any opinion on the part of WRI India concerning the legal status of any country or territory or the delimitation of frontiers or boundaries. Bubble sizes are illustrative, may exceed geographic boundaries and can overlap.

3. Proposed solution

AquaCarbon DC is an AI-assisted decision-support tool for comparing two or three cooling configurations under the same site and workload assumptions. It calculates the expected energy, direct water, carbon and operating-cost outcomes, exposes the assumptions behind each result, and uses Gemini to explain why the trade-offs occur. Users can change a key condition—such as the grid emissions factor or water-availability constraint—and rerun the comparison.

4. The core user experience

Step	What the user does	What AquaCarbon DC returns
1. Configure	Select a representative site context, IT load, reporting period and environmental assumptions.	A clear input summary with source labels and missing-data warnings.
2. Compare	Select two or three cooling profiles and run the scenario.	Side-by-side metrics for energy, PUE, direct water, WUE, carbon and cost where inputs are available.
3. Investigate	Ask why one profile uses less water but more electricity.	A Gemini explanation grounded in calculated values, stated assumptions and identified uncertainties.
4. Test a what-if	Change water availability, weather or the grid emissions factor.	Recalculated results and a clear explanation of which outcomes changed and why.
5. Decide	Review or export the comparison.	A concise decision summary with trade-offs, assumptions, limitations and recommended engineering checks.

5. What makes it different

AquaCarbon DC is not intended to be another PUE dashboard, carbon calculator or general-purpose chatbot. Its differentiator is a site-aware, multi-metric comparison that treats water, energy and carbon as connected constraints. It presents individual metrics and feasibility warnings rather than hiding trade-offs behind one unexplained score.

6. MVP scope

The hackathon MVP will prioritize one representative Indian location context, two cooling profiles as a minimum, and a third only if its model can be supported. The initial demonstration will use sourced environmental context where practical and clearly labelled synthetic facility inputs. It will not depend on access to a real data center.

In scope	Out of scope for the hackathon
Deterministic calculation of energy, PUE, direct water, WUE, carbon and cost when required inputs exist.	Real-time control of chillers, pumps, valves, server fans or other facility equipment.
Visible assumptions, units, sources, formula boundaries and uncertainty labels.	A certified engineering design tool or validated thermal digital twin.
Gemini explanation and bounded what-if questions using calculation outputs.	A claim that modelled potential savings are verified real-world savings.
Cloud deployment and a reproducible scenario dataset.	Full lifecycle assessment, autonomous optimization or comprehensive live telemetry integration.

7. High-level technical approach

Layer	Role	Proposed technology
User interface	Scenario setup, metric comparison, assumptions and AI Q&A.	Web UI; Firebase Hosting or Cloud Run
Application and calculation API	Validate inputs and compute versioned, reproducible metrics.	Python on Cloud Run
Scenario and results data	Store inputs, calculation runs, result objects and provenance.	BigQuery
AI explanation	Explain calculated trade-offs and answer bounded questions.	Gemini on Vertex AI
Environmental context	Provide relevant weather, water-stress or grid-factor inputs when available.	Curated public datasets; optional Earth Engine

Design principle: the calculation engine is the numerical source of truth. Gemini explains the results but must not invent measurements, emission factors or cooling coefficients. Missing data should be shown as unavailable rather than silently replaced with defaults. Any recommendation for a real facility requires qualified engineering review.

8. Success criteria

- A reviewer can compare at least two cooling profiles using the same workload and reporting period.

- All displayed metrics include units, boundaries and traceable assumptions; core calculations pass automated tests.
- A what-if change updates dependent metrics consistently and explains the reason for the change.
- AI responses accurately reflect calculation outputs, mention relevant drawbacks and acknowledge uncertainty.
- The demo runs reliably from a clean session and clearly labels synthetic facility data.
- Environmental impact is reported as modelled potential impact—not as measured savings unless verified before-and-after data exists.

9. Main risks and mitigation

Risk	Mitigation
Generic coefficients could create misleading engineering claims.	Use sourced relationships where available; expose scenario assumptions and ranges; avoid claiming a validated digital twin.
Scope expands beyond the hackathon window.	Build one comparison workflow first; omit Earth Engine and live integrations unless the core experience is complete.
AI produces unsupported numeric claims.	Keep calculations deterministic, pass structured results to Gemini and test numerical fidelity.
Regional water-stress data is mistaken for site-level impact.	Label geographic resolution and distinguish contextual indicators from measured facility water use.

10. Demo narrative

Open with the problem that PUE alone does not capture environmental impact. Configure a representative Bengaluru scenario with clearly labelled illustrative facility inputs, compare two cooling profiles, inspect the energy-water-carbon trade-off, and ask Gemini to explain the result. Change one key assumption and rerun the calculation. Close by showing the sources, limitations and engineering checks required before a real deployment.

Research references

- International Energy Agency, Energy and AI — https://www.iea.org/reports/energy-and-ai/energy-demand-from-ai
- WRI India, More Than Half of India's Data Centres Are in Water-Stressed Regions (18 May 2026) — https://wri-india.org/data/more-half-indias-data-centres-are-water-stressed-regions
- Karimi et al., Water-energy tradeoffs in data centers: A case study in hot-arid climates, Resources, Conservation & Recycling (2022) — https://doi.org/10.1016/j.resconrec.2022.106194
- Google Cloud documentation: BigQuery — https://cloud.google.com/bigquery/docs | Cloud Run — https://cloud.google.com/run/docs | Vertex AI generative AI — https://cloud.google.com/vertex-ai/generative-ai/docs

1. How to optimize for each judging criterion

Technical Merit & GenAI 40%

- Highest priority
- Build a working comparison engine, verified calculation logic, and a Gemini-powered scenario analyst that explains results and trade-offs using actual model outputs.
- Proof for judges: live calculations, working AI interaction, test cases, and a visible Google Cloud integration.

Problem Alignment & Impact 25%

- Demonstrate measurable value
- Show how cooling decisions affect electricity use, water consumption, carbon emissions, operating costs, and thermal safety in a water-stressed location.
- Proof for judges: a baseline-versus-alternative comparison with explicit assumptions and measurable differences.

Innovation & Creativity 25%

- Your differentiation opportunity

- Move beyond conventional efficiency comparisons. Incorporate local water stress and changing operating conditions, then identify when a cooling strategy's environmental advantages come with energy, cost, or reliability trade-offs.
- Proof for judges: a water-stress-aware recommendation that changes when the scenario changes.

User Experience & Design 10%

- Make the value immediately visible
- Provide a simple input form, comparable results, charts, an explanation panel, and a way to change assumptions without needing to understand the underlying engineering.
- Proof for judges: a complete end-to-end demo without manual intervention or notebook execution.

Make the GenAI implementation meaningful

This is the most important architectural decision for a strong technical submission.

Component	Responsibility
Deterministic calculation engine	Produces numeric energy, water, emissions, and cost estimates.
Gemini	Interprets results, answers scenario questions, compares trade-offs, and generates the decision report.
Grounding and validation layer	Supplies Gemini with calculated outputs, source metadata, constraints, and units; validates its structured response.
User interface	Presents results, assumptions, evidence, and AI explanations together.

Do not ask Gemini to invent engineering estimates from a prompt. Use Gemini to reason over results produced by your calculation engine. It should not make unsupported claims about cooling capacity, thermal limits, or water savings.

A compelling GenAI feature is a constraint-aware scenario planner: the user defines limits such as maximum cooling cost, maximum electricity use, and a minimum water-reduction target. The system evaluates the feasible configurations and asks Gemini to explain the best available trade-off. If none meets all constraints, it should say so instead of manufacturing a recommendation.